Fisher forecasts from cached derivatives

How the derivatives are computed and stored, and why every FoM, error and shift is recomputed in seconds

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1 Summary

The question. The project's Fisher notebooks report figures of merit (FoMs), parameter errors and parameter shifts for four bias models, up to four linear-bias priors, seven scale cuts and four bin layouts, under dozens of assumed truths. One evaluation of the data vector runs CAMB, the code that computes the matter power spectrum, and CosmoLike, the code that projects it onto the sky for every thin bin; it takes seconds to tens of seconds. How, then, can a change of model, prior or scale cut cost only seconds?

The answer.

1. A Fisher forecast only needs the derivatives of the data vector at one point (Section 2).
2. For each layout the notebooks compute the base once: one derivative column for each elementary knob, which is each cosmological parameter, each velocity dispersion, and each bias term in each thin bin. They store it on disk (Section 3).
3. A derivative needs more numerical accuracy than a χ^2, and a convergence study fixes the settings every published number uses (Section 4).
4. A bias model is a recipe that moves several knobs at once. Written as a matrix M, it turns the base B into the model's derivatives, $\mathsf{D} = \mathsf{BM}$. A new model is a new M (Section 5).
5. A scale cut keeps some rows of the data vector. Each cut is reduced once to a small matrix G (Section 6), and every Fisher matrix, FoM and shift is then a product of small matrices (Section 7).
6. New evaluations are needed only for a term outside the base, a new fiducial point, or effects beyond first order (Section 8). Section 9 lists the files and what makes them stale.

The setup. The data are the DESI band powers C_ℓ^{gg}: the angular power spectrum of galaxy clustering, averaged in 53 bands of multipole ℓ. They are computed without the Limber approximation, which ties each multipole to a single wavenumber and fails for thin bins, for six tracer groups g (BGS, LRG1, LRG2, LRG3, ELG1 and ELG2) on four layouts: bins of equal comoving width $\Delta d_{\rm com} = 62$ Mpc, and bins of fixed redshift width $\Delta z = 0.02$, 0.05 and 0.2. Bin i has mean redshift $\bar{z}_i$, and $D(z)$ is the linear growth factor, $D(0) = 1$. The bias models and priors are those of the correlated-prior note, `notes/correlated_bias_prior_v2`. Models A and B are two one-loop bias models; model B adds a radial counterterm and widens two priors. The four linear-bias priors (growth-tied, baseline, pessimistic and free) range from one amplitude per group to one free bias per bin; the baseline and pessimistic priors allow correlated wiggles of the bin biases around a straight line in redshift.

2 Why derivatives at one point are enough

Let $m(p)$ be the expected data vector at the parameters p, and p_0 the fiducial point, the parameter values around which the forecast is made. The vector m has n entries: the band powers of every bin's auto spectrum and of every measured cross spectrum between two bins, after 795 entries that the data format reserves for cosmic-shear blocks and that stay zero for DESI alone. C is the covariance of the data: how the measured band powers scatter around m and how their errors correlate.

Near p_0 the data vector is close to a straight line in the parameters,

$$m(p) \;\simeq\; m(p_0) + \mathsf{D}\,(p - p_0), \qquad \mathsf{D}_{rj} = \left.\frac{\partial m_r}{\partial p_j}\right|_{p_0}, \tag{1}$$

with r one row, that is one band power. With Gaussian data errors and Gaussian priors, the probability of the parameters given the data is then a Gaussian in p, whose inverse covariance is the Fisher matrix

$$F \;=\; \mathsf{D}^{\mathsf{T}}\mathsf{C}^{-1}\mathsf{D} \;+\; F_{\text{prior}}. \tag{2}$$

F_{prior} is the priors' contribution: a Gaussian prior of width w adds $1/w^2$ to its parameter's diagonal entry, and a flat prior adds nothing.

The inverse F^{-1} is the forecast covariance of the parameters. $\sigma_j = [(F^{-1})_{jj}]^{1/2}$ is the marginalized error of parameter j, its error when every other parameter is free. $\text{FoM}(w_0, w_a) = [\det \mathsf{C}_w]^{-1/2}$ is the inverse area of the (w_0, w_a) error ellipse, with C_w the (w_0, w_a) block of F^{-1} and $w(a) = w_0 + w_a(1 - a)$ the dark-energy equation of state (a the scale factor). $\text{FoM}(\Omega_m, 10^9 A_s)$ is defined the same way, with Ω_m the matter density today in units of the critical density and A_s the amplitude of the primordial curvature power spectrum.

The model enters F only through D, at one point. The covariance is Gaussian, computed at the one-loop fiducial (the fiducial point of the one-loop bias model, Section 3.4) for 14,000 square degrees, with its inverse rescaled by $f_A = 12{,}300/14{,}000$ to the survey area. It does not depend on the model being fitted.

3 The base

3.1 What the base is

The base is the list of elementary knobs the code turns by hand:

- each cosmological parameter: $10^9 A_s$, the spectral index n_s, the Hubble constant H_0, the baryon density Ω_b, Ω_m, w_0 and w_a;
- each of the seven velocity dispersions $\sigma_{v,1}, \ldots, \sigma_{v,7}$, one per set of bins that shares a value. They smooth each bin's redshift distribution for the random velocities of galaxies (the Fingers-of-God effect);
- separately in every thin bin $i = 1, \ldots, N$: the linear bias $b_{1,i}$; the quadratic and tidal biases $b_{2,i}$ and $b_{s^2,i}$, the galaxies' responses to the squared density and to the tidal field; the counterterms $\alpha_{0,i}$ and $\alpha_{2,i}$, terms proportional to k^2 that absorb small-scale physics; and the stochastic amplitudes $\text{SN}_{0,i}$ and $\text{SN}_{2,i}$, noise-like terms from the discreteness of galaxies.

layout	bins N	cross spectra	entries n	n_{base}	kept, shipped mask
$\Delta d_{\text{com}} = 62$ Mpc	76	97	9,964	547	5,309
$\Delta z = 0.02$	90	133	12,614	645	7,621
$\Delta z = 0.05$	36	52	5,459	267	2,982
$\Delta z = 0.2$	10	13	2,014	85	744

Table 1: The sizes of one layout's Fisher problem. Cross spectra: the pairs of bins whose cross spectrum is in the data vector. n: entries of the data vector. $n_{\text{base}} = 7N + 15$: columns of the base. Last column: the band powers the shipped mask keeps (the mask of the data files, before any scale cut), the largest set of rows any cut uses.

For each knob, the code computes how every band power changes when that knob alone moves: one derivative column, a vector of n numbers. These are the base columns, and they are what the notebooks precompute and store (Section 3.6).

The base matrix B puts the base columns side by side. A model parameter, such as a group offset, a tilt or a per-bin amplitude, is a recipe that moves several knobs at once. The chain rule turns that recipe into a column of the model's derivative matrix: how m changes with p_j is the sum, over the knobs, of how m changes with the knob times how the knob changes with p_j,

$$\frac{\partial m}{\partial p_j} = \sum_a \frac{\partial m}{\partial \kappa_a}\,\frac{\partial \kappa_a}{\partial p_j}, \qquad \text{that is} \qquad \mathsf{D} = \mathsf{B}\,\mathsf{M}, \qquad \mathsf{M}_{aj} = \frac{\partial \kappa_a}{\partial p_j}, \tag{3}$$

with κ_a the value of knob a and p_j the model's parameter j. M is the recipes written as numbers: M_{aj} is how much knob a moves per unit of parameter j. Section 3.2 works one out.

There is no single b_1. Every bias knob belongs to one bin. "The b_1 columns" are therefore N columns, $\partial m/\partial b_{1,1}, \ldots, \partial m/\partial b_{1,N}$, and the same holds for each of the seven bias families $\{b_{1,i}\}$, $\{b_{2,i}\}$, $\{b_{s^2,i}\}$, $\{\alpha_{0,i}\}$, $\{\alpha_{2,i}\}$, $\{\mathrm{SN}_{0,i}\}$ and $\{\mathrm{SN}_{2,i}\}$, $i = 1, \ldots, N$. In order, B has

$$n_{\text{base}} = \underbrace{7}_{\text{cosmology}} + \underbrace{7}_{\sigma_v} + \underbrace{7N}_{7\ \text{families} \times N\ \text{bins}} + \underbrace{1}_{\text{check}} = 7N + 15 \tag{4}$$

columns; the check column is explained in Section 8. On $\Delta d_{\text{com}} = 62$ Mpc there are $N = 76$ bins, so B has $7 + 7 + 532 + 1 = 547$ columns of $n = 9{,}964$ numbers each. Table 1 gives every layout.

3.2 A toy example

The setup. Take one group g of $N = 3$ bins, with mean redshifts $\bar{z}_1$, $\bar{z}_2$, $\bar{z}_3$, their mean $\bar{z}_g$, and $D_i \equiv D(\bar{z}_i)$. Keep one band of each auto spectrum and of the two cross spectra of neighboring bins, so the data vector has $n = 5$ entries,

$$m = \left(C^{11}, C^{22}, C^{33}, C^{12}, C^{23}\right)^{\mathsf{T}}. \tag{5}$$

Keep one cosmological parameter θ and two bias families. The knobs are θ, $b_{1,1}$, $b_{1,2}$, $b_{1,3}$, $b_{2,1}$, $b_{2,2}$ and $b_{2,3}$, so $n_{\text{base}} = 7$.

The base matrix. Name the base columns $y = \partial m/\partial\theta$, $u_i = \partial m/\partial b_{1,i}$ and $v_i = \partial m/\partial b_{2,i}$. Then

$$\mathsf{B} = \begin{array}{c} \\ C^{11} \\ C^{22} \\ C^{33} \\ C^{12} \\ C^{23} \end{array} \begin{array}{c} \begin{array}{ccccccc} y & u_1 & u_2 & u_3 & v_1 & v_2 & v_3 \end{array} \\ \begin{pmatrix} * & * & 0 & 0 & * & 0 & 0 \\ * & 0 & * & 0 & 0 & * & 0 \\ * & 0 & 0 & * & 0 & 0 & * \\ * & * & * & 0 & * & * & 0 \\ * & 0 & * & * & 0 & * & * \end{pmatrix} \end{array}, \tag{6}$$

where $*$ marks a nonzero entry. The column u_1 vanishes on C^{22}, C^{33} and C^{23}, because the bias of bin 1 does not enter those spectra. A cross spectrum, such as C^{12}, depends on the biases of both its bins.

The model. The model has the parameters $p = (\theta, \Delta_{2,g}, t_{2,g}, \beta_1, \beta_2, \beta_3)$, so $n_{\mathrm{par}} = 6$, and sets the knobs by two recipes:

$$b_{1,i} = \frac{\beta_i}{D_i}, \qquad b_{2,i} = b_2^{\mathrm{L}}(b_{1,i}) + \Delta_{2,g} + t_{2,g}\,\zeta_i, \qquad \zeta_i \equiv \bar{z}_i - \bar{z}_g. \tag{7}$$

Here β_i is the amplitude of bin i, and b_2^{L} a calibration: a fitted relation, from simulations, that predicts $b_{2,i}$ from $b_{1,i}$ (the Lazeyras et al. fit [1]). $\Delta_{2,g}$ is the group's offset from that calibration and $t_{2,g}$ its tilt in redshift. Let $s_i \equiv b_2^{\mathrm{L}\prime}(b_{1,i})$ be the slope of the calibration.

The recipes as numbers. M_{aj} is how much knob a moves per unit of parameter j:

1. θ moves only θ. (y is computed by moving θ with the amplitudes β_i held fixed, which is how the model holds them, so θ needs no chain rule.)
2. $\Delta_{2,g}$ moves $b_{2,1}$, $b_{2,2}$ and $b_{2,3}$ by 1 each.
3. $t_{2,g}$ moves $b_{2,i}$ by ζ_i.
4. β_i moves $b_{1,i}$ by $1/D_i$ and, through the calibration, $b_{2,i}$ by s_i/D_i; the other bins do not move.

Each item is one column of

$$\mathsf{M} = \begin{array}{c} \\ \theta \\ b_{1,1} \\ b_{1,2} \\ b_{1,3} \\ b_{2,1} \\ b_{2,2} \\ b_{2,3} \end{array} \begin{array}{c} \begin{array}{cccccc} \theta & \Delta_{2,g} & t_{2,g} & \beta_1 & \beta_2 & \beta_3 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/D_1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/D_2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/D_3 \\ 0 & 1 & \zeta_1 & s_1/D_1 & 0 & 0 \\ 0 & 1 & \zeta_2 & 0 & s_2/D_2 & 0 \\ 0 & 1 & \zeta_3 & 0 & 0 & s_3/D_3 \end{pmatrix} \end{array}. \tag{8}$$

The multiplication. Column j of $\mathsf{D} = \mathsf{BM}$ is the sum of the columns of B, weighted by column j of M:

$$\frac{\partial m}{\partial \theta} = y, \qquad \frac{\partial m}{\partial \Delta_{2,g}} = v_1 + v_2 + v_3,$$
$$\frac{\partial m}{\partial t_{2,g}} = \zeta_1 v_1 + \zeta_2 v_2 + \zeta_3 v_3, \qquad \frac{\partial m}{\partial \beta_i} = \frac{u_i + s_i v_i}{D_i}. \tag{9}$$

Written out entry by entry, the offset column is

$$\frac{\partial m}{\partial \Delta_{2,g}} = \left(\frac{\partial C^{11}}{\partial b_{2,1}}, \frac{\partial C^{22}}{\partial b_{2,2}}, \frac{\partial C^{33}}{\partial b_{2,3}}, \frac{\partial C^{12}}{\partial b_{2,1}} + \frac{\partial C^{12}}{\partial b_{2,2}}, \frac{\partial C^{23}}{\partial b_{2,2}} + \frac{\partial C^{23}}{\partial b_{2,3}} \right)^{\mathsf{T}} : \tag{10}$$

each auto spectrum responds to its own bin's $b_{2,i}$, and each cross spectrum to both of its bins'.

Another model, the same base. The free-per-bin model replaces the group offset and tilt by one offset per bin, $b_{2,i} = b_2^{\mathrm{L}}(b_{1,i}) + \Delta_{2,i}$. Each offset moves one knob, so its block on the $b_{2,i}$ rows is the 3×3 identity:

$$\mathsf{M}_{\mathrm{free}} = \begin{array}{c} \\ \theta \\ b_{1,1} \\ b_{1,2} \\ b_{1,3} \\ b_{2,1} \\ b_{2,2} \\ b_{2,3} \end{array} \begin{array}{c} \begin{array}{ccccccc} \theta & \Delta_{2,1} & \Delta_{2,2} & \Delta_{2,3} & \beta_1 & \beta_2 & \beta_3 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/D_1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/D_2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1/D_3 \\ 0 & 1 & 0 & 0 & s_1/D_1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & s_2/D_2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & s_3/D_3 \end{pmatrix} \end{array}, \qquad \frac{\partial m}{\partial \Delta_{2,i}} = v_i. \tag{11}$$

The priors.

- Group model: $1/w_g^2$ on $\Delta_{2,g}$ (a Gaussian prior of width w_g), nothing on $t_{2,g}$ (a flat prior).
- Free per bin: $1/w_g^2$ on each $\Delta_{2,i}$.
- Free per bin, $\sqrt{N}$-scaled: the same $\mathsf{M}_{\mathrm{free}}$, with $1/(3\,w_g^2)$ on each $\Delta_{2,i}$. The mean of the three offsets then has the width $w_g\sqrt{3}/\sqrt{3} = w_g$ of the group offset.

Three models, one B: the group and free models differ in M, the free model and its $\sqrt{N}$-scaled version only in F_{prior}. The base, the expensive part, never changes.

3.3 How a derivative column is computed

A finite difference. The code cannot differentiate CAMB and CosmoLike by hand. It moves one knob by a small step ϵ, recomputes the data vector, and divides the change by the step. The simplest such finite difference, the central difference $[m(\theta_0 + \epsilon) - m(\theta_0 - \epsilon)]/(2\epsilon)$, is off by a term of order ϵ^2 times the third derivative of m: the curvature it ignores.

The five-point stencil. A stencil is the set of points at which m is evaluated, with the weights that combine them. The code uses five points, $\theta_0 \pm \epsilon$, $\theta_0 \pm 2\epsilon$ and θ_0, with the weights

$$\frac{\partial m}{\partial \theta} \simeq \frac{m(\theta_0 - 2\epsilon) - 8\, m(\theta_0 - \epsilon) + 8\, m(\theta_0 + \epsilon) - m(\theta_0 + 2\epsilon)}{12\, \epsilon}. \tag{12}$$

These weights cancel the error terms through fourth order, so the remaining error falls as ϵ^4, and the formula is exact for any dependence up to quartic. The center point has weight zero and is never evaluated, so one column costs four evaluations.

The cosmological and σ_v columns. These are stencils of sampled parameters of the model A likelihood, taken by `model.derivatives` of the Fisher module (`interface/cosmolike_lssty1xdesi_fisher.py`). The step is relative, $\epsilon = h\,\theta_0$ with $h = 0.02$; a parameter whose fiducial value is zero (w_a) takes an absolute step, scaled by its prior width when its prior is Gaussian. Each evaluation is a full call of the likelihood, CAMB included, with the cache of previous results of cobaya (the framework that runs the likelihood) disabled, so that no stale vector is reused.

The same call also returns the stencils of the six group amplitudes B_g, which set the linear bias of every bin of group g at the fiducial point, $b_{1,i} = B_g / D(\bar{z}_i)$. They enter no Fisher matrix of the model A and B notebooks, because the per-bin columns replace them; they are kept for the chain-rule check of Section 8.

3.4 The per-bin columns, in batches

In model A the real-space galaxy power spectrum of bin i is

$$P_{gg,i}(k) \;=\; b_{1,i}^2\, P_{\rm mm}(k) \;+\; P_{\rm loop}(k;\, b_{1,i}, b_{2,i}, b_{s^2,i}) \;+\; \alpha_{0,i}\, k^2 P_{\rm mm}(k), \tag{13}$$

plus the stochastic terms of Section 3.5. $P_{\rm mm}$ is the nonlinear matter power spectrum. $P_{\rm loop}$ holds the one-loop terms, the next order of perturbation theory beyond linear, which carry $b_{2,i}$ and $b_{s^2,i}$. $\alpha_{0,i}\, k^2 P_{\rm mm}$ is a counterterm, in $h^{-2}\,{\rm Mpc}^2$: a term that absorbs small-scale physics the model does not compute.

None of the four families $\{b_{1,i}\}$, $\{b_{2,i}\}$, $\{b_{s^2,i}\}$ and $\{\alpha_{0,i}\}$ enters m linearly, so each needs a stencil.

The stencil of one bin. For the linear bias of bin i, with every other knob at its fiducial value:

1. evaluate m four times, with $b_{1,i}$ replaced by $b_{1,i}(1 \pm \epsilon)$ and $b_{1,i}(1 \pm 2\epsilon)$;
2. combine the four vectors with the weights of Eq. (12), the step being $\epsilon\, b_{1,i}$:

$$\frac{\partial m}{\partial b_{1,i}} \simeq \frac{m\big(b_{1,i}(1-2\epsilon)\big) - 8\, m\big(b_{1,i}(1-\epsilon)\big) + 8\, m\big(b_{1,i}(1+\epsilon)\big) - m\big(b_{1,i}(1+2\epsilon)\big)}{12\, \epsilon\, b_{1,i}}. \tag{14}$$

The step of $b_{1,i}$ is relative, a fraction $\epsilon = 0.01$ of $b_{1,i}$. The steps of the other three are absolute: the four points are $b_{2,i} \pm \epsilon$ and $b_{2,i} \pm 2\epsilon$ with $\epsilon = 0.02$, the same for $b_{s^2,i}$, and $\epsilon = 5\, h^{-2}\,{\rm Mpc}^2$ for $\alpha_{0,i}$.

spectrum	$\partial/\partial b_{1,1}$	$\partial/\partial b_{1,2}$	$\partial/\partial b_{1,3}$	$\partial/\partial b_{1,4}$	$\partial/\partial b_{1,5}$
C^{11}	A	0	0	0	0
C^{22}	0	B	0	0	0
C^{33}	0	0	A	0	0
C^{44}	0	0	0	B	0
C^{55}	0	0	0	0	A
C^{12}	A	B	0	0	0
C^{23}	0	B	A	0	0
C^{34}	0	0	A	B	0
C^{45}	0	0	0	B	A

Table 2: Five bins in a row, cross spectra between neighbors only. Each entry names the batch whose evaluations fill it: batch A moves bins 1, 3 and 5, batch B moves bins 2 and 4. A 0 is exactly zero, because the spectrum does not contain that bin.

The problem: one bin at a time is wasteful. Done bin by bin, a family costs 4 evaluations per bin: $4 \times 76 = 304$ on $\Delta d_{\rm com} = 62$ Mpc, and 1,216 for the four families. Most of that work is wasted. The data vector of that layout holds 173 spectra, 76 autos and 97 crosses, and bin i enters at most five of them: its own auto spectrum and up to four cross spectra. Every other row of $\partial m/\partial b_{1,i}$ is exactly zero, and an evaluation that moves one bin leaves 168 or more spectra unchanged. Counted per spectrum, the block of one family, 173 spectra by 76 columns, is 98% zeros; in the cached arrays no column has a nonzero entry outside its own bin's spectra.

The idea: move several bins in one evaluation. Take five bins in a row whose only measured cross spectra join neighbors: 1–2, 2–3, 3–4 and 4–5. Put the bins in two batches, sets of bins moved together in one stencil:

- batch A moves $b_{1,i}$ in bins 1, 3 and 5 together;
- batch B moves $b_{1,i}$ in bins 2 and 4 together.

Each batch is one five-point stencil, four evaluations. Table 2 shows which batch fills each entry of the five columns $\partial m/\partial b_{1,1}, \ldots, \partial m/\partial b_{1,5}$.

Reading the table:

- An auto spectrum C^{ii} is filled in the batch that moves bin i.
- A cross spectrum C^{ij} appears in two columns, filled from two different batches: $\partial C^{ij}/\partial b_{1,i}$ from the batch where bin i moves and bin j stays put, and $\partial C^{ij}/\partial b_{1,j}$ from the batch where j moves and i stays put.
- A 0 is not computed: a spectrum that does not contain bin i cannot depend on $b_{1,i}$, so the code sets those rows to zero.

Bin by bin, the five columns would cost $5 \times 4 = 20$ evaluations; the two batches cost $2 \times 4 = 8$.

A batch is a set of bins, not of spectra. No batch holds "the autos" or "the crosses". In the evaluations of the batch that contains bin i, bin i's auto spectrum and all of its cross spectra receive their derivative with respect to $b_{1,i}$. A cross spectrum C^{ij} is visited twice, once in the batch of each of its two bins. Figure 2c shows this on a real block.

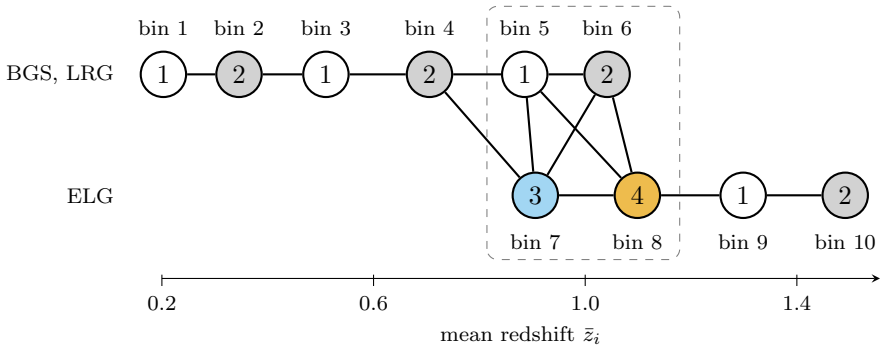


Figure 1: The cross-pair graph of the $\Delta z = 0.2$ layout, all ten bins. Each dot is a bin at its mean redshift (top row BGS and LRG, bottom row ELG), and a line joins two bins whose cross spectrum is in the data vector. The number in each dot is its batch from the greedy rule, repeated by the dot's shade. Bins 5 to 8 (dashed box: two LRG3 and two ELG1 bins, which overlap in redshift) are all joined to each other, so they need four different batches.

When may two bins move together? Why not move every bin at once and keep, for each column, only its own rows?

- That works for the auto spectra: C^{ii} depends only on bin i's biases.
- It fails for the cross spectra. If bins i and j move together by δ, C^{ij} changes by $(\partial C^{ij}/\partial b_{1,i} + \partial C^{ij}/\partial b_{1,j})\,\delta$, and one evaluation cannot split the two terms.

So two bins may share a batch only if no measured spectrum contains both. Batching is "move as many bins at once as possible, then keep each column's own rows", with the fewest batches this rule allows.

What a batch is, as a picture. Draw each bin as a dot, and join two dots with a line when the data vector contains their cross spectrum: this is the cross-pair graph (Figure 1).

- A batch is a set of dots with no line between any two of them, so its bins can move together.
- Batching the bins means giving every dot a batch number, so that two joined dots never share a number.
- The code (`pair_coloring`) uses a greedy rule: it goes through the bins in data-vector order and gives each the lowest batch number that none of its already numbered neighbors has.

Mathematicians call such batches colors, from the picture of labelling the dots so that joined dots never share a label; hence the name of the code's function.

The real layout, step by step. Take $\Delta d_{\rm com} = 62$ Mpc. Its $N = 76$ bins are numbered 1 to 76 in data-vector order, by group and, within a group, by redshift (numbers start at 1, as everywhere in this note and in Figure 2; the code counts from 0). DESI's galaxies come in six groups, ordered by redshift (Table 3). Figure 2 draws the whole layout and its batches; the steps below walk through it.

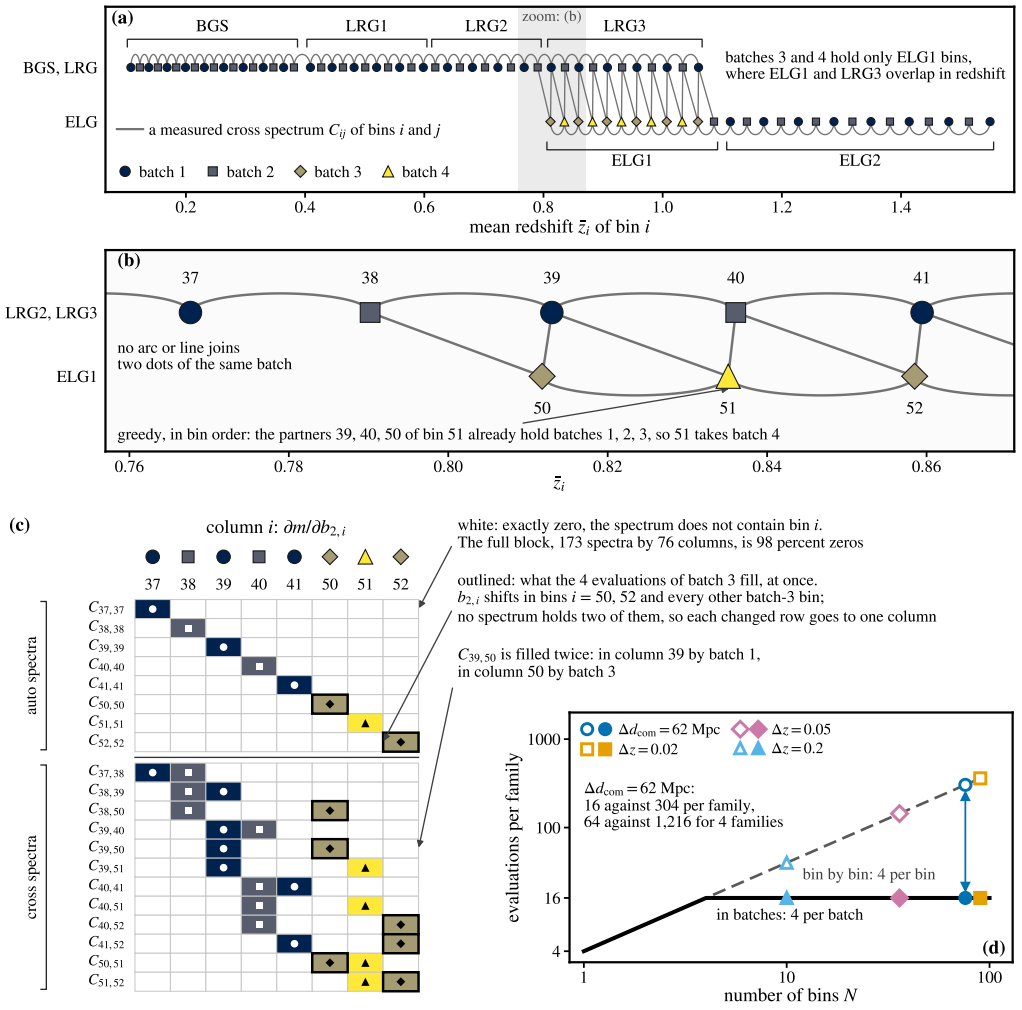


Figure 2: How the per-bin bias columns of $\Delta d_{\rm com} = 62$ Mpc are computed in batches. (a) Every bin at its mean redshift, BGS and LRG on the top row, ELG on the bottom; an arc or line is a measured cross spectrum, and the marker gives the batch. Look first at the long chains: each bin is crossed only with its neighbors, so the markers alternate between batch 1 (circle) and batch 2 (square). Then look at the shaded strip, where LRG3 and ELG1 cover the same redshifts: the short lines between the rows are their cross spectra, and only there do batches 3 (diamond) and 4 (triangle) appear. (b) A zoom on that strip. No line joins two markers of the same batch. Follow the greedy rule at bin 51: its partners 39, 40 and 50 already hold batches 1, 2 and 3, so it takes batch 4. (c) The block $\partial m/\partial b_{2,i}$ of those eight bins, with a row per spectrum and a column per bin. A column is nonzero only on the spectra that contain its bin; white cells are exactly zero. The outlined cells are what the four evaluations of batch 3 fill at once. A cross spectrum such as $C_{39,50}$ is filled twice: in column 39 by batch 1, and in column 50 by batch 3. (d) Evaluations per bias family against the number of bins N: bin by bin the cost grows as $4N$ (open markers, dashed line); in batches it stops at $4 \times 4 = 16$ (filled markers), whatever N.

group	galaxies	bins	mean redshifts $\bar{z}_i$
BGS	bright galaxies	1–18	0.11–0.38
LRG1	luminous red galaxies	19–29	0.41–0.60
LRG2	luminous red galaxies	30–38	0.62–0.79
LRG3	luminous red galaxies	39–49	0.81–1.06
ELG1	emission-line galaxies	50–61	0.81–1.09
ELG2	emission-line galaxies	62–76	1.11–1.55

Table 3: The six groups of $\Delta d_{\rm com} = 62$ Mpc. LRG3 and ELG1 cover the same redshifts: they overlap.

Which cross spectra are measured. The rule is simple: the data vector holds the cross spectrum of two bins when their redshift distributions $n(z)$ overlap or touch, that is, when the bins are neighbors in redshift. Bins far apart in redshift share no spectrum. For the fixed-Δz layouts this is the rule of `cross_pairs` in `scripts/make_desi_dz_case.py`; on $\Delta d_{\rm com} = 62$ Mpc the pair list comes from the DESI forecast, and the same rule reproduces 96 of its 97 pairs (the exception is the BGS–LRG1 boundary pair, bins 18 and 19).

On $\Delta d_{\rm com} = 62$ Mpc the 97 cross spectra come in two kinds:

1. 74 join a bin to the next one along the chain of bins: neighboring thin shells, which touch at their shared edge. These include the four links at group boundaries: BGS–LRG1, LRG1–LRG2, LRG2–LRG3 and ELG1–ELG2.

2. 23 join an LRG bin to an ELG1 bin at nearly the same redshift: 22 LRG3–ELG1 pairs and one LRG2–ELG1 pair at $z \approx 0.8$. This is the overlap, explained below.

Each bin has 1 to 4 cross partners.

Along the chain: batches 1 and 2. The greedy rule walks through the bins in order. Bin 1 takes batch 1. Bin 2 is crossed with bin 1, so it takes batch 2. Bin 3 is crossed with bin 2 only, among the bins already numbered, so it takes batch 1 again, and so on: along the chain the bins alternate.

- The evaluations of batch 1 move bins $1, 3, 5, \ldots$ They fill $\partial C^{11}/\partial b_{1,1}$, $\partial C^{33}/\partial b_{1,3}$, $\ldots$, and the cross spectra C^{12}, C^{23}, C^{34} through their odd end: $\partial C^{12}/\partial b_{1,1}$, $\partial C^{23}/\partial b_{1,3}$, $\partial C^{34}/\partial b_{1,3}$.

- The evaluations of batch 2 move bins $2, 4, 6, \ldots$ and fill the even ends: $\partial C^{12}/\partial b_{1,2}$, $\partial C^{23}/\partial b_{1,2}$, $\partial C^{34}/\partial b_{1,4}$, and the even auto spectra.

This is the five-bin example of Table 2, continued through every group: the alternating circles and squares of Figure 2a.

The overlap: why batches 3 and 4 appear. LRG3 and ELG1 are two different galaxy populations, luminous red galaxies and emission-line galaxies, observed in the same volume, at $z \approx 0.8$–1.1 (Table 3). An LRG3 bin and an ELG1 bin at the same redshift trace the same matter, so their cross spectrum is measured, roughly $b_{1,i}\, b_{1,j}\, P$ with P the matter power

spectrum. It carries no shot-noise term, because the two bins hold different galaxies: the $\mathrm{SN}_{0,i}$ cross terms exist only within one sample.

These pairs obey the same rule: the two bins are neighbors in redshift. They only look far apart because the bin numbers run group by group. LRG3 bin 44 ($\bar{z} = 0.932$) and ELG1 bin 55 ($\bar{z} = 0.931$) are 11 apart in number but at the same redshift, and their cross spectrum is measured. In Figure 1, drawn against redshift, the lines between the LRG and ELG rows are short.

For batching this matters because an ELG1 bin's redshift neighbors are its ELG neighbors and two LRG bins at nearly the same redshift. Bin 51, for example, is crossed with bins 50 and 52 (ELG1) and with bins 39 and 40 (LRG3). Bins 39 and 40 already sit in batches 1 and 2, and bin 50 in batch 3, so bin 51 takes batch 4. The greedy rule puts the eleven ELG1 bins 50 to 60 in batches 3 (bins $50, 52, \ldots, 60$) and 4 (bins $51, 53, \ldots, 59$); batch 1 holds 33 bins and batch 2 holds 32, and the last ELG1 bin, 61, lands in batch 2. Batches 3 and 4 appear only in the overlap (Figure 2b).

Four batches, whatever the number of bins. Every bin's partners are local in redshift, so the number of batches does not grow with N. The other layouts follow the same pattern, neighbors along the chain plus overlap pairs: $\Delta z = 0.02$ has 133 cross spectra (88 along the chain) and 1 to 5 partners per bin; $\Delta z = 0.05$ has 52 (34) and 1 to 5; $\Delta z = 0.2$ has 13 (8) and 1 to 4. All four layouts get four batches.

A rule of thumb explains why the number stays small: if every bin's partners lie within k places of it in redshift order, $k + 1$ batches suffice, because going through the bins in that order, each bin has at most k numbered partners to avoid. The partners reach 2 places away on $\Delta d_{\mathrm{com}} = 62$ Mpc and 3 places on the three fixed-Δz layouts. Those three need four batches, the minimum: each contains four bins all crossed with each other, two LRG3 and two ELG1 bins at the same redshift (bins 36, 37, 51 and 52 on $\Delta z = 0.02$; 15, 16, 21 and 22 on $\Delta z = 0.05$; 5 to 8 on $\Delta z = 0.2$, Figure 1). On $\Delta d_{\mathrm{com}} = 62$ Mpc no four bins are all crossed with each other, only three, such as bins 39, 50 and 51, and three batches would suffice; the greedy rule, which goes through the bins in data-vector order, all LRG bins before the ELG bins, gives four.

On every layout batches 3 and 4 hold only ELG1 bins. On $\Delta d_{\mathrm{com}} = 62$ Mpc they are 11 of the 76 bins, about 15%, yet they take half of each family's evaluations.

What batching buys. The cost now follows from the number of batches, not from N. On $\Delta d_{\mathrm{com}} = 62$ Mpc:

1. one family takes 4 batches $\times$ 4 evaluations = 16 evaluations and fills all 76 of its columns, instead of $4 \times 76 = 304$;

2. the four stenciled families take $4 \times 16 = 64$ evaluations, instead of 1,216.

The same 64 hold on every layout, from 10 bins to 90 (Figure 2d).

Why not exact derivatives by hand? If each cross spectrum were a fixed bilinear expression in the biases, its derivatives could be written down exactly. The low-rank non-Limber engine of CosmoLike does not compute it that way. For each bin it approximates $\sqrt{P_{g,i}(k,z)}$, with $P_{g,i}$ the one-loop galaxy power spectrum of bin i, by a short sum of products of a function

of k and a function of z: a truncated singular-value decomposition, which makes the projection fast.

It builds a cross spectrum from the geometric mean $\sqrt{P_{g,i}}\sqrt{P_{g,j}}$ of the two bins' spectra. Before the square root it floors $P_{g,i}$ at $10^{-15}\, b_{1,i}^2 P_\delta$, with P_δ the matter power spectrum, so the root always exists. The square root, the truncation and the floor make the spectra not exactly bilinear in the biases. Finite differences through the full likelihood differentiate the model exactly as it is computed.

Why the call to CosmoLike is intercepted. The sampled parameters offer no per-bin handle: the parameter files tie each bin's linear bias to its group amplitude, and the likelihood computes $b_{2,i}$ and $b_{s^2,i}$ from $b_{1,i}$ through calibrations. The likelihood also sets every value at every evaluation, so a value written into CosmoLike beforehand would be overwritten.

The per-bin values exist separately in one place only: the setter, the function `set_nuisance_bias_tidal` through which the likelihood hands them to CosmoLike's C code. The notebook replaces it by a wrapper, its own function that changes the values of one batch and then calls the original. The original is restored after the steps.

The loop in the code. `model_a_derivatives` of the model A notebook does exactly this:

1. The wrapper records the per-bin arrays the likelihood passes: `B1`, `B2`, `BS2` and `BK`, one entry per bin. In a batch evaluation it changes the entries of the bins of that batch. `B1` is multiplied by $1 + o\,\epsilon$; `B2` and `BS2` get $o\,\epsilon$ added; an $\alpha_{0,i}$ step $o\,\epsilon$ enters `BK` as $o\,\epsilon/[2\, b_{1,i}(c/H_0)^2]$, because CosmoLike's k^2 term is $2\, b_{1,i} b_{K,i}\, k^2 P_{\rm mm}$ with k in units of H_0/c. Here o runs over the stencil offsets $(2, 1, -1, -2)$ (`fm.FIVE_POINT_OFFSETS`).

2. For each family and each of the four batches it evaluates the full data vector at the four offsets and combines them with `fm.five_point`, Eq. (12).

3. Column i of the family's block gets that derivative on the rows of every spectrum that contains bin i, and zero elsewhere. The $b_{1,i}$ column is divided by $b_{1,i}$, because its step is relative.

3.5 Exact templates

A template is the exact derivative of a term that is linear in its amplitude: the term itself, per unit amplitude. Three families are linear:

- the white noise beyond Poisson $\mathrm{SN}_{0,i}$, in units of the bin's Poisson power $1/\bar{n}_i$ ($\bar{n}_i$ its tracer density);

- the stochastic term $\mathrm{SN}_{2,i}\, k^2\mu^2$, with μ the cosine of the angle to the line of sight, in units of $f_{\rm sat}\sigma_{\rm sat}^2/\bar{n}_i$ ($f_{\rm sat}$ the satellite fraction, $\sigma_{\rm sat}$ the satellites' velocity dispersion in distance units);

- the radial counterterm $\alpha_{2,i}$ of model B, in $h^{-2}\,\mathrm{Mpc}^2$, which enters the counterterm density $\frac{1}{2}(b_{1,i}\alpha_{0,i} + f\mu^2\alpha_{2,i})\, k^2\delta_{\rm L}$ (f the linear growth rate, $\delta_{\rm L}$ the linear density contrast).

array	shape	what it is
`EXAMPLE_EVALUATE_FISHER_MODEL_A_DERIVATIVES_1.pkl` (32 MB)		
`m0`	9,964	the fiducial band powers $m(p_0)$
`D`	$9{,}964 \times 20$	stencil columns: 7 cosmology, 7 σ_v, 6 B_g (check only)
`ops['b1']`	$9{,}964 \times 76$	column i: $\partial m/\partial b_{1,i}$, the linear bias of bin i alone
`ops['b2']`	$9{,}964 \times 76$	column i: $\partial m/\partial b_{2,i}$
`ops['bs2']`	$9{,}964 \times 76$	column i: $\partial m/\partial b_{s^2,i}$
`ops['a0']`	$9{,}964 \times 76$	column i: $\partial m/\partial \alpha_{0,i}$, per $h^{-2}\,\mathrm{Mpc}^2$
`T`	$9{,}964 \times 76$	column i: $\partial m/\partial \mathrm{SN}_{0,i}$
`b1`, `b2`, `bs2`	76 each	the fiducial $b_{1,i}$, $b_{2,i}$, $b_{s^2,i}$ of every bin
`EXAMPLE_EVALUATE_FISHER_MODEL_A_SN2_1.pkl` (12 MB)		
`T2`	$9{,}964 \times 76$	column i: $\partial m/\partial \mathrm{SN}_{2,i}$
`EXAMPLE_EVALUATE_FISHER_MODEL_B_TEMPLATES_1.pkl` (18 MB)		
`a2`	$9{,}964 \times 76$	column i: $\partial m/\partial \alpha_{2,i}$, per $h^{-2}\,\mathrm{Mpc}^2$

Table 4: The precomputed arrays of $\Delta d_{\mathrm{com}} = 62$ Mpc, in Python's pickle files: $n = 9{,}964$ band powers and $N = 76$ bins. Each $9{,}964 \times 76$ block is 6.1 MB in double precision. The files also hold the settings they were made with, and the second and third repeat `m0` and the $\partial m/\partial \mathrm{SN}_{0,i}$ columns for the checks of Section 8. In the toy's notation, `ops['b1']` is $[u_1, \ldots, u_N]$ and `ops['b2']` is $[v_1, \ldots, v_N]$.

The likelihood projects the $\mathrm{SN}_{0,i}$ and $\mathrm{SN}_{2,i}$ terms directly (`stochastic_templates` of `likelihood/desi_cl.py`, orders 0 and 2). It uses each bin's kernel $W_i(\chi)$: the bin's redshift distribution as a function of comoving distance χ, after the Fingers-of-God smoothing. $\mathrm{SN}_{0,i}$ projects through $\int d\chi\, W_i W_j/\chi^2$, and $\mathrm{SN}_{2,i}$ through $\int d\chi\, W_i' W_j'/\chi^2$, with $W' = dW/d\chi$, because $k^2\mu^2 = k_\parallel^2$ is a second derivative along the line of sight (the correlated-prior note, section "Model A"). These templates need only the kernels of one fiducial evaluation.

$\alpha_{2,i}$ has its own per-bin setter, `set_nuisance_bias_alpha2`, wrapped the same way. A central difference on each batch gives its columns, exact because the term is linear: two evaluations per batch, eight in all.

3.6 What is precomputed, and what is computed at every run

The work splits into three levels.

1. **On disk, once per layout (hours):** the base columns and the fiducial vector $m(p_0)$. Table 4 lists every array on $\Delta d_{\mathrm{com}} = 62$ Mpc.

2. **In memory, at every run (seconds):** for each scale cut m, its kept rows K_m and the matrix $\mathsf{G}_m = f_A\, \mathsf{B}_K^{\mathsf{T}} \mathsf{C}_{KK}^{-1} \mathsf{B}_K$, 547×547 on $\Delta d_{\mathrm{com}} = 62$ Mpc (Section 6). The covariance is read fresh from its file at every run.

3. **For every model and prior (a fraction of a second):** M, then $F = \mathsf{M}^{\mathsf{T}} \mathsf{G}_m \mathsf{M} + F_{\mathrm{prior}}$, its inverse, the FoMs, the shifts and $\langle \chi^2 \rangle$ (Sections 6 and 7). No evaluation of the data vector at all.

What makes the disk part stale. Only what changes the knobs' responses: the fiducial point, the likelihood settings (accuracy, options, parameter files) and the code. The fingerprint

term	what it is	evaluations
1	the fiducial vector	1
$4\,N_{\mathrm{stencil}}$	20 stencil parameters: 7 cosmology, 7 σ_v, 6 B_g	80
$4\,N_{\mathrm{batch}}\,N_{\mathrm{fam}}$	4 batches, 4 families $\{b_{1,i}\}$, $\{b_{2,i}\}$, $\{b_{s^2,i}\}$, $\{\alpha_{0,i}\}$	64
$2\,N_{\mathrm{batch}} + 2$	$\{\alpha_{2,i}\}$: 2 per batch, plus a fiducial vector before and after	10
1	the fiducial run of the $\{\mathrm{SN}_{2,i}\}$ templates	1
	total	156

Table 5: The evaluations behind one layout's base. The $\{\mathrm{SN}_{0,i}\}$ and $\{\mathrm{SN}_{2,i}\}$ templates cost no evaluation beyond a fiducial one.

configuration	grid factor g	seconds per evaluation
MCMC settings	32	3.9
reference, grid lowered	64	11.4
reference	128	17.0
reference, grid raised	256	28.0
reference, grid raised	512	48.4
candidate final configuration	256	19.5

Table 6: The cost of one data-vector evaluation on $\Delta z = 0.05$, measured by the convergence study on AC power with no other heavy job running, except the $g = 256$ reference run, which started while one other job was running. The candidate final configuration ran 130 evaluations in 42 minutes. Section 4 describes the configurations.

of Section 4 catches changes of the input; a change of the code needs a flag set by hand (Section 9). The recipe M, the priors, the scale cut and the covariance file do not make it stale.

3.7 How many data-vector evaluations, and how long

Counted in the builders and the Fisher module, the base of one layout takes

$$N_{\mathrm{eval}} \;=\; 1 + 4\,N_{\mathrm{stencil}} + 4\,N_{\mathrm{batch}}\,N_{\mathrm{fam}} + (2\,N_{\mathrm{batch}} + 2) + 1 \tag{15}$$

evaluations of the data vector, term by term in Table 5. It is 156 on every layout, because no term grows with the number of bins.

The cost of one evaluation. Each evaluation is a complete data vector: CAMB, then CosmoLike's non-Limber projection of every thin bin. Its cost depends mostly on g, the refinement factor of the comoving-distance grid on which the non-Limber projection integrates (Section 4). Table 6 gives the times the convergence study measured on $\Delta z = 0.05$ (36 bins), on AC power. The MCMC settings are those of the example yamls used for Markov-chain Monte Carlo runs, the sampling of the full likelihood.

The total time. The base of a layout takes $N_{\mathrm{eval}} \times t_{\mathrm{eval}}$, with t_{eval} the time of one evaluation. At the candidate final configuration:

- $\Delta z = 0.05$, from the measured t_{eval}: 156×19.5 s ≈ 51 minutes.

- The other layouts, a rough estimate that scales t_{eval} with the number of bins: $\Delta d_{\mathrm{com}} = 62$ Mpc (76 bins) 156×41 s ≈ 1.8 hours; $\Delta z = 0.02$ (90 bins) 156×49 s ≈ 2.1 hours; $\Delta z = 0.2$ (10 bins) 156×5.4 s ≈ 14 minutes.

The four layouts then take about five hours. The recompute at the final settings will replace these estimates by measured times (Section 4).

4 Derivative accuracy

Why a derivative needs more than a χ^2. Every evaluation of m carries numerical noise: the integration grids, the non-Limber distance grid and CAMB's tolerances change the vector by small amounts that do not vary smoothly with the parameters. A finite difference divides that noise by the step, so a setting that converges a χ^2 (the MCMC criterion is $|\Delta\chi^2| < 0.2$) can leave a derivative noisy.

The noise biases the forecast one way. A column $d_j + e_j$, with e_j its noise, gives $F_{jj} = (d_j + e_j)^{\mathsf{T}}\mathsf{C}^{-1}(d_j + e_j)$, whose average exceeds the true value by $\langle e_j^{\mathsf{T}}\mathsf{C}^{-1}e_j\rangle > 0$: noise looks like information, and the FoM comes out too high.

The convergence study, `tests/studies/run_fisher_derivatives.py`, measures this on $\Delta z = 0.05$ for the paper model, model B + tilts on the pessimistic prior. It scores the marginalized errors of w_0, w_a, Ω_m and A_s, both FoMs, and the shifts of the break-test truths (truths that carry a term the fit lacks or holds with a prior; the correlated-prior note, "Model B and the reach of model A"), at $k_{\max} = 0.075$ and $0.2\,h$/Mpc. The MCMC settings raise $\mathrm{FoM}(w_0, w_a)$ by 57% at $0.075\,h$/Mpc and by 10% at $0.2\,h$/Mpc, although their fiducial vector differs from the reference by only $\Delta\chi^2 = 0.015$.

The reference, and one control at a time. The reference configuration sets every accuracy control high: CosmoLike's `accuracyboost` 4 and `integration_accuracy` 2, the low-rank engine's rank 8 (the number of terms kept in its sum) and multipole cap 2000, a grid factor $g = 128$, and CAMB's `AccuracyBoost` 2.0, `k_per_logint` 30 and maximum wavenumber $80\,\mathrm{Mpc}^{-1}$. Here g is the product `accuracyboost` $\times$ `nonlimber_accuracyboost`, and the non-Limber projection integrates over $511\,g+1$ comoving distances; the MCMC value on $\Delta z = 0.05$ is $g = 32$. Each other configuration lowers one control, and g is also raised to 256 and 512.

The noise of one evaluation. Through five equally spaced stencil points, the combination $m(-2) - 4m(-1) + 6m(0) - 4m(1) + m(2)$ of a smooth vector is of order ϵ^4 times its fourth derivative, negligible at these steps, while noise independent from point to point gives it $1 + 16 + 36 + 16 + 1 = 70$ times the variance of one point. The study whitens it, multiplying by L^{-1} with $\mathsf{C} = LL^{\mathsf{T}}$, so that every entry counts in units of the data errors, and divides its length by $\sqrt{70}$. The result is the noise of one evaluation in $\sqrt{\chi^2}$ units (Table 7).

At the MCMC settings the Ω_m, w_0 and w_a columns are seven to twelve times noisier than the H_0 column, which does not change with the grid. The finer grid quiets w_0 and w_a most: the w_a noise falls fortyfold from the MCMC settings to $g = 256$.

The distance grid dominates. Table 8 gives the offset of $\mathrm{FoM}(w_0, w_a)$ from its value at $g = 256$, every other control at the reference. Swapping single columns between configurations

configuration	H_0	Ω_m	w_0	w_a
MCMC settings	2.1×10^{-3}	1.7×10^{-2}	2.5×10^{-2}	1.4×10^{-2}
$g = 64$	2.2×10^{-3}	8.1×10^{-3}	1.1×10^{-2}	4.8×10^{-3}
$g = 128$ (reference)	2.2×10^{-3}	6.5×10^{-3}	4.3×10^{-3}	1.8×10^{-3}
$g = 256$	2.2×10^{-3}	6.1×10^{-3}	2.2×10^{-3}	3.5×10^{-4}

Table 7: The noise of one evaluation along each stencil parameter, in $\sqrt{\chi^2}$ units, on $\Delta z = 0.05$. The rows with a grid factor g keep every other control at the reference.

grid factor g	$k_{\max} = 0.075\,h/\mathrm{Mpc}$	$k_{\max} = 0.2\,h/\mathrm{Mpc}$
32 (MCMC)	+38%	+6.7%
64 (old caches)	+31%	+5.1%
128	+0.9%	+0.3%

Table 8: The offset of FoM(w_0, w_a) of the paper model from its value at $g = 256$, on $\Delta z = 0.05$, every other control at the reference. $k_{\max}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

shows where the offset comes from: the w_a column carries it, then w_0 and Ω_m; the per-bin and σ_v columns move the FoM by less than 0.1%.

The criterion: independence of the step. The truncation error of a stencil grows with the step, as ϵ^4 for five points, while the noise it divides falls as $1/\epsilon$. A derivative is converged when neither matters: when the FoMs, the errors and the shifts stay put as the step changes, every step scaled together, and as the stencil changes from three points to five.

At $g = 128$ the five-point stencil at $h/2$ raises the FoM by 18%: the noise divided by a smaller step. At $g = 256$ the FoM is flat within about ±1% (FoM$(\Omega_m, 10^9 A_s)$ within ±1.4%) from $h/2$ to $3h$, and the three-point stencil agrees with the five-point one. The tolerance on the FoMs is about 1 to 2%; $g = 128$ agrees with $g = 256$ to 0.9% only at $h = 0.02$, so it stays a documented cross-check, and the final tier uses $g = 256$ on this layout.

The CAMB halofit patch. Halofit is CAMB's fitting formula for the nonlinear matter power spectrum. It finds the nonlinear scale by bisection (repeated halving of an interval) on $\sigma(R)$, the rms linear density contrast smoothed on the scale R; stock CAMB stops at $|\sigma(R) - 1| \leq 10^{-3}$. Where the bisection stops jumps between nearby cosmologies and shifts the nonlinear power spectrum by about 0.3%, a noise that a finite difference divides by its step.

Cocoa's installation key `PATCH_CAMB_HALOFIT_TOLERANCE`, on by default, tightens the test to 10^{-7}. The final tier requires it: the builders stop at start unless `halofit.f90` holds the 10^{-7} test and the compiled CAMB library is newer than that file (`check_halofit_patch`).

Two tiers. The project fixes two sets of accuracy settings, two tiers, which `derivative_settings` returns per layout.

- **Exploration**: fast settings for quick looks; its FoMs are not converged.
- **Final**: the settings the study converged, for every figure and number of a paper or of a note's final version. The criterion is the stability of the FoMs, the errors and the shifts; runtime is not a criterion.

Both tiers use five-point stencils for every stencil and per-bin column, and exact templates for $\{\mathrm{SN}_{0,i}\}$, $\{\mathrm{SN}_{2,i}\}$ and $\{\alpha_{2,i}\}$. The caches on disk today are the exploration tier, made with central differences for the four per-bin families (on $\Delta z = 0.05$ they gave $\mathrm{FoM}(w_0, w_a) = 0.227$ at $0.075\,h/\mathrm{Mpc}$, against 0.157 converged), and the builders refuse them until the final-tier recompute.

Each cache records its tier, its settings, and a fingerprint (`settings_fingerprint`): a 64-character code (a sha256 hash) computed from the full model input, so that any change of the input changes the code. The input is the example yaml merged with the likelihood's options and parameter files, the accuracy settings, the priors and the ties between parameters, together with the fiducial point and the tier. Unlike a comparison of file names, the fingerprint changes when a yaml is edited under the same name, and stays the same when a file only moves.

TODO, still being measured.

- The full final settings of each layout: CosmoLike's `accuracyboost` and `integration_accuracy`, the low-rank rank and multipole cap, and CAMB's `AccuracyBoost`, k sampling and maximum wavenumber.
- The step check on $\Delta d_{\mathrm{com}} = 62$ Mpc at $g = 256$.
- The comparison with DerivKit, an external package that fits polynomials through many points (Cocoa's `DERIVKIT` keys; `get_ddv_dkit`).
- The agreement of the production route, the wrapped setter (Section 3.4), with the study's route, which unties the per-bin parameters and steps them through the likelihood.
- The measured time of the final-tier recompute on each layout, to replace the estimates of Section 3.7.

5 The models of the notebooks as matrices

The toy's recipes carry over to the real models, with seven bias families instead of two and six groups (BGS, LRG1, LRG2, LRG3, ELG1, ELG2) instead of one. Every model is a matrix M of Eq. (3) on the same B, written down without any evaluation (`variant_columns` in the model B notebook). Figure 3 draws its structure. Its columns:

- **Cosmology and σ_v**: unit vectors, as θ in the toy.
- **A group offset**: ones on the rows of the group's bins in its family's block, as $\Delta_{2,g}$ in the toy. The offsets $\Delta_{2,g}$ and $\Delta_{s,g}$ sit on the $b_{2,i}$ and $b_{s^2,i}$ blocks, and the group's $\alpha_{0,g}$, $\alpha_{2,g}$, $\mathrm{SN}_{0,g}$ and $\mathrm{SN}_{2,g}$ on the $\alpha_{0,i}$, $\alpha_{2,i}$, $\mathrm{SN}_{0,i}$ and $\mathrm{SN}_{2,i}$ blocks.
- **A tilt**: the weights $\bar{z}_i - \bar{z}_g$ on the same rows, as $t_{2,g}$ in the toy. Model A + tilts tilts $\Delta_{2,g}$, $\Delta_{s,g}$ and $\alpha_{0,g}$, and model B + tilts also $\alpha_{2,g}$, in every group of more than one bin, with flat priors; $\mathrm{SN}_{0,g}$ and $\mathrm{SN}_{2,g}$ never tilt.
- **The per-bin amplitude**: as β_i in the toy, plus the tidal bias, which follows $b_{1,i}$ by co-evolution (the relation for galaxies that form with no tidal bias and then move with the matter), $b_{s^2,i} = -\frac{4}{7}(b_{1,i} - 1) + \Delta_{s,g}$. The column has three entries:

$$\frac{\partial m}{\partial \beta_i} = \frac{1}{D(\bar{z}_i)} \left[\frac{\partial m}{\partial b_{1,i}} + b_2^{\mathrm{L}\prime}(b_{1,i}) \frac{\partial m}{\partial b_{2,i}} - \frac{4}{7} \frac{\partial m}{\partial b_{s^2,i}} \right]. \tag{16}$$

- **The growth-tied prior**: one amplitude per group, $b_{1,i} = B_g/D(\bar{z}_i)$, whose column is the sum of the group's β_i columns.
- **Free per bin, and $\sqrt{N}$-scaled**: as in the toy, for all six families, on the free linear-bias prior; the widths are the group parameter's w_g, or $w_g\sqrt{N_g}$ with N_g the number of bins in the group.

The four variants of the notebooks are four matrices M on one B: model A (one $\Delta_{2,g}$, $\Delta_{s,g}$, $\alpha_{0,g}$, $\mathrm{SN}_{0,g}$ and $\mathrm{SN}_{2,g}$ per group, nothing on the $\alpha_{2,i}$ block), model A + tilts, model B (adds one $\alpha_{2,g}$ per group and widens the offset priors) and model B + tilts. On $\Delta d_{\mathrm{com}} = 62$ Mpc, model A has $n_{\mathrm{par}} = 14+5\times6+76 = 120$ parameters (the cosmology and σ_v, five families in six groups, one amplitude per bin), and model B + tilts $14+6\times6+4\times6+76 = 150$. Switching between them, or to any other model built from the same knobs, needs no evaluation of the data vector.

6 A scale cut is a set of rows

A scale cut removes the band powers that small, nonlinear scales dominate. A cut at $k_{\max}$, quoted at $z = 0$, cuts bin i at $k_{\max}/D(\bar{z}_i)$, by the non-Limber ultraviolet-fraction criterion of Saraivanov et al. [2] within the shipped mask: a band is kept while less than 5% of it comes from the power spectrum above that wavenumber. The Fisher matrix at a cut follows in four steps.

1. The cut keeps a set K_m of rows of m. It drops the other rows of B and keeps every column: a cut never changes a derivative. B_K is the kept rows of B, and C_{KK} the kept block of the covariance.
2. The data part of the Fisher matrix of a model at this cut is $F_{\mathrm{data}} = \mathsf{D}_K^{\mathsf{T}} f_A \mathsf{C}_{KK}^{-1} \mathsf{D}_K$. Since $\mathsf{D}_K = \mathsf{B}_K\mathsf{M}$,
$$F_{\mathrm{data}} \;=\; \mathsf{M}^{\mathsf{T}}\big(f_A\,\mathsf{B}_K^{\mathsf{T}}\,\mathsf{C}_{KK}^{-1}\,\mathsf{B}_K\big)\mathsf{M} \;=\; \mathsf{M}^{\mathsf{T}}\,\mathsf{G}_m\,\mathsf{M}, \qquad \mathsf{G}_m \;\equiv\; f_A\,\mathsf{B}_K^{\mathsf{T}}\,\mathsf{C}_{KK}^{-1}\,\mathsf{B}_K. \tag{17}$$
G_m is what the data of cut m say about every pair of knobs: its entry (a, b) is the information shared by knobs a and b.
3. G_m contains neither M nor a prior. It depends only on the base, the cut and the covariance, so the notebooks compute it once per cut, and every model and prior at that cut reuses it.
4. The priors add their precision: $F = \mathsf{M}^{\mathsf{T}}\mathsf{G}_m\mathsf{M} + F_{\mathrm{prior}}$.

The toy, cut. Suppose cut m drops the cross spectrum C^{23}, which fails the ultraviolet criterion. Then:

- $K = \{C^{11}, C^{22}, C^{33}, C^{12}\}$, and B_K is the 4×7 matrix of the first four rows of Eq. (6);
- C_{KK} is 4×4, and G_m is 7×7;
- the group model's F is $(6\times7)(7\times7)(7\times6) = 6\times6$, and the free model's 7×7, both from the same G_m.

After this cut, v_3 survives only on C^{33}.

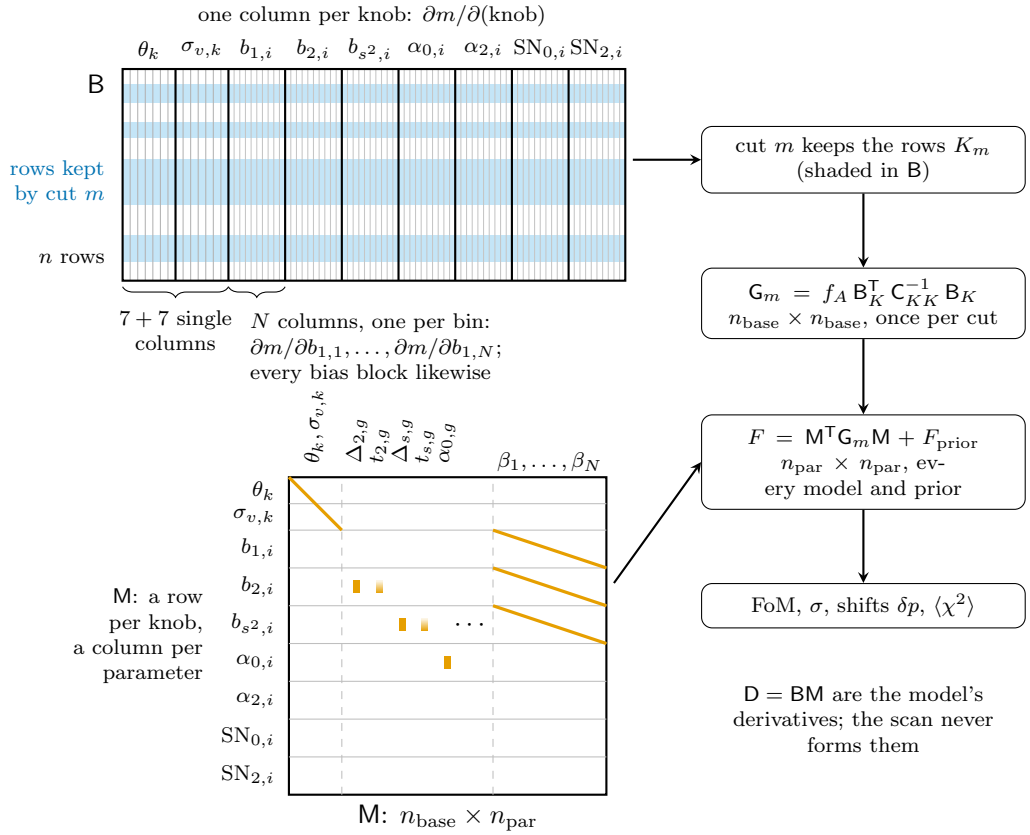


Figure 3: How the notebooks assemble a Fisher matrix. Top left: the base B of one layout. Each cosmological parameter θ_k and each velocity dispersion $\sigma_{v,k}$ has one column; every bias family is a block of N columns, one per thin bin (thin gray lines). Shaded: the rows a cut m keeps. Bottom left: a model's recipe matrix M, with a row per knob, in the column order of B, and a column per model parameter. Orange marks its nonzero entries: the identity on the cosmology and σ_v; for one group g, the offset $\Delta_{2,g}$ (ones on the group's bins in the $b_{2,i}$ block), its tilt $t_{2,g}$ (the weights $\bar{z}_i - \bar{z}_g$, drawn as a gradient), and likewise $\Delta_{s,g}$, $t_{s,g}$ and $\alpha_{0,g}$; and three diagonals for the per-bin amplitudes β_i, Eq. (16). Right: a cut is reduced once to G_m, and every model and prior at that cut is $\mathsf{M}^{\mathsf{T}}\mathsf{G}_m\mathsf{M}$ plus priors. The check column of B is not drawn.

The real sizes. On $\Delta d_{\mathrm{com}} = 62$ Mpc, B is $9{,}964 \times 547$. The shipped mask keeps 5,309 rows, so B_K is $5{,}309 \times 547$, C_{KK} is $5{,}309 \times 5{,}309$ and G_m is 547×547. F is 120×120 for model A and 150×150 for model B + tilts. The large block is handled once per cut, and the 12 Fisher systems at that cut (four variants, three priors; a Fisher system is one Fisher matrix with its inverse and results) reuse G_m.

How G_m is computed. The notebooks compute G_m in `layout_grams`, without ever inverting the covariance:

1. they write $\mathsf{C}_{KK} = LL^{\mathsf{T}}$ with L lower triangular, the Cholesky factorization, a matrix analogue of the square root;

2. they solve $L\,W = \mathsf{B}_K$ for W row by row, which a triangular L allows (forward substitution);

3. they form $\mathsf{G}_m = f_A\,W^{\mathsf{T}}W$.

This saves work, avoids amplifying round-off when the covariance is ill-conditioned (when its inverse magnifies small errors), and makes G_m symmetric with no negative information, by construction.

Each layout gets eight such matrices: the seven cuts $k_{\max} = 0.05, 0.075, \ldots, 0.2\,h/\mathrm{Mpc}$, and the shipped mask alone, the reference of the information test below. The model A notebook, which works at one cut, forms $\mathsf{D} = \mathsf{BM}$ explicitly and multiplies it by the inverse of the kept covariance block; the two routes agree (Section 8).

The priors. F_{prior} adds precision and needs no data. The diagonal Gaussian priors, all centered on zero, sit on σ_v (from the example yaml) and on the model's group parameters. The counterterm widths, $0.5/k_{\max}^2$, change with the cut, so the diagonal is rebuilt for each cut (`prior_diagonal`).

The linear-bias prior acts on the block of the β_i. The baseline and pessimistic priors add, group by group, the penalty of their memory model on bin-to-bin wiggles of the linear biases (`group_bias_precision`; the correlated-prior note, sections "What the prior charges" and "How the prior enters the Fisher matrix"). The free prior adds nothing, and the growth-tied prior is a change of M.

Parameters without data. At small cuts some bins keep no band, and their columns vanish on K_m. A parameter whose information at the cut, data plus prior, falls below 10^{-10} of its information on the shipped mask with the same priors leaves the fit; a Gaussian prior keeps it in. A cut that keeps no band (as $\Delta z = 0.02$ at $k_{\max} = 0.05\,h/\mathrm{Mpc}$) or leaves a cosmological parameter without data gives no fit.

The pseudo-inverse. A plain inverse fails in one more case. On $\Delta d_{\mathrm{com}} = 62$ Mpc at $k_{\max} = 0.075\,h/\mathrm{Mpc}$ only 7 of the 18 BGS bins keep a band. Under model A + tilts on the free prior, the flat-prior BGS tilts then leave one combination of parameters with no information at all. Along such a direction a plain inverse amplifies round-off by about 10^{16}; it once produced a spurious 2σ shift.

`stable_inverse` therefore takes a pseudo-inverse, an inverse that skips the directions with no information:

1. it scales F to unit diagonal, $\tilde{F} = SFS$ with $S = \mathrm{diag}(F)^{-1/2}$, so that the parameters' units do not matter;

2. it diagonalizes $\tilde{F} = V\Lambda V^{\mathsf{T}}$: the columns of V are the parameter combinations measured independently, and the eigenvalues in Λ the information in each;

3. it drops the directions whose eigenvalue is below 10^{-12} of the largest, and returns

$$F^{-1} \;=\; S\,V_\kappa\,\Lambda_\kappa^{-1}\,V_\kappa^{\mathsf{T}}\,S, \tag{18}$$

with κ the kept directions.

A dropped direction must leave (w_0, w_a), Ω_m and A_s alone: the scan gives no fit when one touches them by more than 10^{-6}.

7 Everything downstream reuses G

Once G_m is built, nothing in the notebooks touches an $n \times n$ matrix or the data vector again. F is Eq. (17) plus the priors, inverted by Eq. (18), and σ_j and the FoMs follow as in Section 2.

Shifts. A truth is an assumed true universe that departs from the fitted model, for example one with $\alpha_{2,i} \neq 0$ when the fit has no $\alpha_{2,i}$. It moves the data by δm. When δm lies in the span of the base columns, $\delta m = \mathsf{B}c$, with c a vector of n_{base} coefficients:

- $\alpha_{2,i} = 12.5\,h^{-2}\,\text{Mpc}^2$ in every bin i is $c = 12.5$ on the $\alpha_{2,i}$ block;
- an offset truth puts its values on the $b_{2,i}$ and $b_{s^2,i}$ blocks;
- a fractional drift δ_i of the linear biases is $c = \sum_i \beta_i\,\delta_i\,\mathsf{M}_{\beta_i}$, with M_{β_i} the column of Eq. (16).

To first order the best fit then moves by

$$\delta p \;=\; F^{-1}\mathsf{D}^{\mathsf{T}}\mathsf{C}^{-1}\delta m \;=\; F^{-1}\mathsf{M}^{\mathsf{T}}\,\mathsf{G}_m\,c \tag{19}$$

(the correlated-prior note, "The shift formula"). $\delta m = \mathsf{B}c$ is exact for the $\alpha_{2,i}$, $\text{SN}_{0,i}$ and $\text{SN}_{2,i}$ truths, whose terms are linear, and first order for truths in $b_{1,i}$, $b_{2,i}$ and $b_{s^2,i}$, since the band powers are quadratic in them. A truth costs one product with G_m.

Random truths. Eq. (19) is linear in c, so a family of random truths collapses into one matrix: for drifts δ_i of the linear biases the (w_0, w_a) shift is $\delta w = A\,\delta$, with A the (w_0, w_a) rows of Eq. (19) applied to the N columns $\beta_i\mathsf{M}_{\beta_i}$. Two thousand truths cost one matrix product.

The exact $\langle\chi^2\rangle$. Whether a fit's error bars are honest is tested by averaging $\chi^2_w = (\hat{w} - w_{\text{true}})^{\mathsf{T}}\mathsf{C}_w^{-1}(\hat{w} - w_{\text{true}})$, with $\hat{w}$ the fitted (w_0, w_a), over the truths and the data noise. Every source is Gaussian and enters linearly, so the average is exact in closed form (the correlated-prior note, "The test statistic"):

$$\langle\chi^2_w\rangle \;=\; \Delta^2_{\text{drift}} \;+\; \text{tr}\Big[\mathsf{C}_w^{-1}\left(A\,\Sigma\,A^{\mathsf{T}} + R\,\Sigma_\theta\,R^{\mathsf{T}} + N_w\right)\Big]\,. \tag{20}$$

- Δ_{drift} is the (w_0, w_a) shift, in units of the ellipse, of the known drift alone: the smooth change of the linear biases with redshift that physics predicts.
- Σ is the covariance of the truth's remaining scatter of the linear biases.
- R and Σ_θ are the response to, and the prior variance of, any other parameter drawn at random ($\alpha_{2,g}$ per group in the random test of model B).
- N_w is the (w_0, w_a) block of $F^{-1}F_0F^{-1}$, the share of the data noise and the Gaussian priors, with F_0 the Fisher matrix without the linear-bias prior.

The notebooks report $\sqrt{\langle\chi^2\rangle/n_{\text{dof}}}$, with $n_{\text{dof}} = 2$ for (w_0, w_a), and $n_{\text{dof}} = 1$ for Ω_m or A_s with that parameter's own row; 1 means honest error bars. The random draws only make the percentile bars of the figures.

The cost. Each G_m costs one Cholesky factorization of at most 7,621 rows, about $n_K^3/3 \approx 1.5 \times 10^{11}$ floating-point operations, and one triangular solve, eight times per layout. A Fisher system then costs $\mathsf{M}^\mathsf{T}\mathsf{G}_m\mathsf{M}$, with $n_{\rm base} \le 645$ and $n_{\rm par} \le 164$ (model B + tilts on $\Delta z = 0.02$), about $2\,n_{\rm base}^2 n_{\rm par} \approx 1.4 \times 10^8$ operations, plus the diagonalization of an $n_{\rm par} \times n_{\rm par}$ matrix: a fraction of a second.

The $k_{\rm max}$ scan of the model B notebook has 336 systems (four layouts, seven cuts, four variants, and the baseline, pessimistic and free priors; the growth-tied prior fails the accuracy test and is left out), 324 with a fit and 12 without any band. In operations, all of them together cost less than the Cholesky factorizations of one layout. Without the base, each variant would need its own derivatives, at the cost of Section 3.7 per layout, and each truth its own data vector.

8 Limits and checks

What needs new evaluations of the data vector.

- **A physical term outside the base.** A term that is no combination of the knobs, such as a nonzero third-order bias or redshift-space one-loop terms (which CosmoLike lacks), needs new base columns. A new combination of existing knobs, such as a quadratic tilt or another calibration slope at the same fiducial values, is only a new M.

- **A different fiducial point.** Every derivative is taken at p_0. A new fiducial cosmology, new fiducial biases, or a calibration law that moves the fiducial $b_{2,i}$ needs new base columns, and a new covariance, which sits at the one-loop fiducial. A new covariance alone needs no evaluation: the notebooks rebuild every G_m from the current covariance file at every run.

- **Dependence beyond first order.** The forecast is the curvature at p_0, and Eq. (19) is first order in δm. The band powers are quadratic in the biases, so a large truth, such as an offset of twice a prior width or the 22% BGS flux-limit drift, is treated to first order, and a shift near the $\sigma/3$ threshold needs a refit, with new evaluations, to be confirmed.

The checks that guard the machinery.

- **The chain rule.** The per-bin columns, obtained through the wrapped setter and summed over a group, $\sum_{i\in g} \partial m/\partial\beta_i$ with Eq. (16), must equal the stencil derivative $\partial m/\partial B_g$, obtained through the full likelihood. The two routes share no evaluation; they agree to 1.2×10^{-7} of the stencil column's largest entry (the correlated-prior note, "Numerical checks").

- **Bit-for-bit template checks.** The $\partial m/\partial \mathrm{SN}_{0,i}$ templates of a template run must equal the cached ones exactly, and the run's fiducial vector the cached one. B carries the difference of the two vectors as its check column, so their $\Delta\chi^2$ is a diagonal entry of G on the shipped mask; it is zero. After the $\alpha_{2,i}$ steps the vector must equal the one before bit for bit (the setter is back at zero), and the forward and backward $\alpha_{2,i}$ differences must agree (the term is linear).

- **The matched-prior check.** When the truth's linear-bias scatter equals the fit's prior and every other parameter is drawn from the fit's own priors, the scatter and noise terms of Eq. (20) must add up to exactly n_{dof}: the widening the prior gives the ellipse equals the error the scatter causes (the correlated-prior note, "The test statistic"). The notebooks verify this on every system to round-off, which tests G, M, the priors and the pseudo-inverse together. The model A FoMs of the model B notebook's scan also equal those of the model A notebook, assembled by the other route of Section 6, to 7.9×10^{-10}, and a $+$ and a $-$ truth give shifts of equal size.
- **The cache check.** Each cache stores its settings, tier and fingerprint, and a notebook refuses a cache whose entries differ from its own (Section 9).

9 Using the caches

The caches are the pickle files of Table 4, one set per layout, in `chains/`, which git ignores: a fresh clone recomputes them (hours), and every later run reads them. Their names start with `EXAMPLE_EVALUATE_FISHER`, and $\langle n \rangle = 1$ to 4 numbers the layouts in the order $\Delta d_{\mathrm{com}} =$ 62 Mpc, $\Delta z = 0.02$, 0.05, 0.2.

- `_SCALE_CUTS.pkl`, written by the linear notebook, `EXAMPLE_EVALUATE_FISHER.ipynb`: the bands each ultraviolet-fraction cut keeps, at $k_{\mathrm{max}} = 0.05, 0.075, \ldots, 0.25\,h/\mathrm{Mpc}$; the scans use the seven cuts up to 0.2.
- `_MODEL_A_DERIVATIVES_<n>.pkl` and `_MODEL_A_SN2_<n>.pkl`, written by the model A notebook, `EXAMPLE_EVALUATE_FISHER_MODEL_A.ipynb`, with the flags `RECOMPUTE` and `RECOMPUTE_SN2`.
- `_MODEL_B_TEMPLATES_<n>.pkl`, written by the model B notebook, `EXAMPLE_EVALUATE_FISHER_MODEL_B.ipynb`, with the flag `RECOMPUTE_TEMPLATES`.

The model A and B notebooks cache no Fisher matrix, G_m or FoM: these depend on the covariance and are rebuilt at every run.

The flags. With a flag at `False`, the default, a notebook reads an existing cache after the cache check and computes a missing one; `True` recomputes it and overwrites the file. The layouts are computed cheapest first, and each cache is written as soon as its layout is done, so an interrupted run keeps the finished layouts. The model B notebook reads the model A caches and never writes them; it stops when one is missing or was made with other settings. For a fresh clone, run the linear notebook (for the cuts), then the model A notebook, then the model B notebook.

What makes a cache stale.

- **Detected automatically**: any change of the model input that the fingerprint covers (a likelihood option, a CAMB argument, a prior, a tie between parameters, the fiducial point), of the tier, of the stencil rule or of the steps. The notebook stops and asks for the flag. A file that moves or is renamed with the same content keeps its cache.

- **Not detected, so set the flag by hand**: a change of the code itself (the likelihood's Python, CosmoLike, CAMB beyond the halofit check), since the fingerprint covers the input, not the code. No check compares a cache with the current code unless a cache is recomputed; the fiducial-vector check then catches a change that moves the fiducial vector.
- **Needs no recomputation**: a new covariance, a new prior or prior width, a new model built from the cached knobs, and any cut already in the cut cache. A $k_{\max}$ outside it needs the linear notebook to compute that cut (CosmoLike's split of each band into the power below and above the cut), but no new derivative.

References

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[2] E. Saraivanov, H. Rubira, V. Miranda and T. Eifler, "Towards the Two-Loop EFTofLSS in Galaxy Lensing Surveys", 2026, arXiv:2603.13031.